

STUDENT PERFORMANCE REPORT

Power BI Dashboard Analytics & Insights

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Tool: Microsoft Power BI

1. Executive Summary

This report presents a comprehensive analysis of student academic performance using an interactive Power BI dashboard. The dashboard consolidates data across multiple dimensions including parental involvement, family income, internet access, school type, hours studied, and access to resources. The objective is to identify key performance drivers and provide actionable insights to educators, administrators, and policymakers to improve student outcomes.

The analysis reveals that a combination of socioeconomic factors, study habits, and resource availability significantly influences student exam scores. Public school students represent the majority of the dataset (69.57%), and students with internet access and high access to resources consistently demonstrate stronger academic performance.

2. Project Overview

2.1 Objectives

- Analyze the distribution of exam scores across demographic and socioeconomic groups.
- Identify the relationship between study hours, attendance, and academic performance.
- Examine the impact of internet access and resource availability on student outcomes.
- Compare performance between public and private school students.
- Assess the influence of parental involvement and family income on exam results.

2.2 Data Overview

Field	Details
Report Title	Student Performance Report
Tool Used	Microsoft Power BI
Key Metrics	Exam Score, Attendance, Hours Studied, Previous Scores
Dimensions	School Type, Gender, Internet Access, Motivation Level, Resource Access, Parental Involvement, Family Income
School Types	Public (69.57%), Private (30.43%)
Filters Available	Internet Access (Yes/No), Motivation Level (High/Low/Medium)

3. Dashboard Visuals & Analysis

3.1 Exam Score by Parental Involvement, Family Income & Internet Access

This grouped bar chart illustrates the interaction between parental involvement level (High, Low, Medium), family income, and internet access. Key observations include:

- Students from high-income households with high parental involvement show the highest aggregate exam scores, approaching 100K.
- Internet access consistently boosts scores across all income and involvement levels.
- Low parental involvement groups, regardless of income, show reduced exam performance.
- The interplay between family income and internet access suggests a digital equity factor in academic outcomes.

3.2 Exam Score & Attendance vs. Hours Studied

This dual-line chart captures the relationship between weekly study hours and both exam performance and attendance:

- Both exam scores and attendance rise steadily from 0 to approximately 20 hours studied per week.
- Peak performance is observed at around 20 hours of weekly study, suggesting an optimal study threshold.
- Beyond 20 hours, a noticeable decline occurs in both metrics, indicating diminishing returns or student burnout.
- Attendance and exam scores track closely, highlighting the strong positive correlation between presence and academic success.

3.3 Hours Studied, Exam Score & Attendance by Internet Access

This scatter/bubble chart plots exam scores against hours studied, segmented by internet access (Yes/No):

- Students with internet access cluster at higher exam scores and higher study hours, reaching up to 0.3M in aggregate scores.
- Students without internet access are concentrated in the lower-left quadrant, indicating fewer study hours and lower scores.
- The disparity underscores the critical role of internet connectivity in enabling effective self-directed learning.

3.4 Exam Score by School Type

The donut chart presents the proportion of total exam scores attributed to each school type:

School Type	Total Exam Score	Percentage
Public	309K	69.57%
Private	135K	30.43%

- Public school students contribute a significantly larger share of total exam scores, consistent with their larger population representation.
- Private school students, while fewer in number, perform competitively on a per-student basis.

3.5 Exam Score & Previous Scores by Access to Resources

This horizontal stacked bar chart compares exam and previous scores segmented by resource access level (High, Medium, Low):

- High resource access students record the largest combined exam and previous score totals, exceeding 0.4M.
- Medium resource access students perform moderately in the 0.2-0.3M range.
- Low resource access students have the lowest scores, emphasizing the need for targeted resource allocation.
- Consistency between previous and current scores across resource levels confirms resource access as a stable, long-term predictor.

3.6 Exam Score by Hours Studied, Motivation Level, School Type & Gender

The small-multiples area chart breaks down exam scores by study hours across four subgroups: Private Female, Private Male, Public Female, and Public Male, with motivation level overlaid:

- High motivation students consistently outperform low and medium motivation peers across all subgroups.

- Public school students, particularly females, show a broader distribution of high-motivation learners.
- The study-hour sweet spot of 20-30 hours per week is consistent across all gender and school-type groups.
- Private school students show tighter clustering, suggesting more homogeneous study habits within that population.

4. Key Findings

Finding	Detail
Optimal Study Hours	Peak exam performance is achieved at approximately 20 hours per week; beyond this, returns diminish significantly.
Internet Access Gap	Students with internet access score dramatically higher, confirming a digital divide impact on learning.
Parental Involvement	High parental involvement is strongly linked to better exam outcomes, especially combined with higher family income.
Resource Access	Access to learning resources is a consistent, long-term predictor of both previous and current academic performance.
School Type	Public school students dominate in volume; private school students show competitive per-capita performance.
Motivation Level	High motivation is the single strongest within-student predictor of academic success across all demographic groups.
Attendance	Attendance closely tracks with exam scores, reinforcing the importance of consistent school presence.

5. Recommendations

5.1 For Educators & Schools

- Encourage structured study schedules capped at 20-25 hours per week to maximize performance without burnout.
- Implement regular attendance monitoring as an early warning system for at-risk students.
- Use motivation-level segmentation to design differentiated learning support programs.

5.2 For Administrators & Policymakers

- Prioritize internet access programs for students in low-income households to bridge the digital equity gap.

- Expand resource availability (textbooks, tutoring, labs) especially in under-resourced schools and communities.
- Develop family engagement initiatives to raise parental involvement levels across income brackets.

5.3 For Further Analysis

- Conduct per-student (normalized) comparisons between public and private school performance.
- Apply longitudinal tracking of student cohorts to assess whether early resource access sustains achievement.
- Build predictive models on this dataset to identify at-risk students before exam periods.

6. Conclusion

The Student Performance Power BI Dashboard provides a powerful, multi-dimensional view of the factors shaping academic outcomes. The analysis confirms that no single variable drives student success in isolation. A combination of access-related factors (internet, resources, family income), behavioral factors (study hours, attendance), and motivational factors collectively determine performance.

By leveraging these insights, educational institutions can move toward a more targeted, data-driven approach to student support. The recommendations outlined in this report offer a concrete starting point for systemic improvements that can positively impact student outcomes at scale.

The Power BI dashboard, with its interactive filters and cross-dimensional visuals, serves as an ongoing analytical tool that stakeholders can use to monitor trends, evaluate interventions, and drive continuous improvement in educational delivery.